Fiscal Year:	FY 2012	**Task Last Updated:**	FY 10/19/2012
PI Name:	Kennedy, Ann R. D.Sc.		
Project Title:	Center of Acute Radiation Research		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSBRI-RFA-08-02 Center of Acute Radiation Research (CARR)
Start Date:	09/01/2008	**End Date:**	08/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note: title change per NSBRI (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Freund, Gregory (University of Illinois at Urbana-Champaign) Wan, X. (University of Pennsylvania) Weissman, Drew (University of Pennsylvania) Cengel, Keith M.D., Ph.D. (University of Pennsylvania) Ebert, Douglas (Wyle Laboratories)		
Grant/Contract No.:	NCC 9-58-RE01801		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In this research, animal model systems are being utilized to evaluate the space radiation induced acute radiation syndrome (ARS) with vomiting (and/or retching) and white blood cell counts in ferrets, white blood cell counts, fatigue, and immune system parameters in mice, and skin injury, with accompanying immune system changes, and white blood cell counts in pigs as the biological endpoints. In addition, the effects of combined exposure to simulated hypogravity and space radiation on blood cell counts and immune functions with respect to both the innate immune system and the acquired immune system in mice are being evaluated in mice. The first few years of the project are focused on determinations of whether certain parts of the ARS occur in animals exposed to space radiation(s) simulating that occurring during a solar particle event (SPE). If problems are observed to exist for the model systems used, later years of the project will be focused on the development of countermeasures for these effects. In these studies, selected drugs will be evaluated in the animal model systems to determine their effectiveness in prevention and/or mitigation of the specific ARS symptoms. The dose response relationships measured for these endpoints will be compared mathematically with published results for human subjects to determine the RBEs and the predictive value of these animal models for bio-dosimetry and for intervention studies to evaluate FDA-approved drugs that can be used for prevention and/or mitigation of the ARS symptoms. Various tissues will be collected from the animals for histopathological examination and for studies by various immunohistochemistry, qPCR, and/or immunochemistry techniques to study the underlying mechanisms for the acute radiation syndrome and its prevention and/or treatment by drugs. As part of the studies in year 3 of this grant's activities, countermeasure studies began to identify agents that could prevent and/or mitigate the effects of SPE radiation on ferret emesis and mouse blood cell counts. These investigations have been designed to address Acute Radiation Syndromes - Specific Gaps 1, 2, 3, 5, 6, 7 and 8 identified in NSBRI-RFA-08-02.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is well-known that exposure to ionizing radiation at sufficiently high doses can result in various types of adverse biological effects. The countermeasures to be developed as part of this research program have significance for all individuals exposed to high doses of ionizing radiation on Earth as well as in space. The countermeasures shown to protect against radiation induced biological effects from space radiation will be equally useful for protection against radiation induced biological effects on Earth.
Task Progress:	For immune and hematopoietic cells, significant decreases in white blood cell counts were observed in mice and ferrets irradiated at high and low dose rates, starting at doses of 25-50 cGy and up to 2 Gy. For vomiting and retching, increases were observed in ferrets irradiated at high and low dose rates, starting at doses of 75 cGy and up to 2 Gy. For total skin irradiation, there are significant wound-healing problems following skin irradiation (abnormal wound healing was observed for some biopsy sites during the 5 week observation period). For total skin irradiation, at increasing doses of radiation, skin becomes more sensitive to touch (lymphedematous), and there can be blistering, burns, and epithelial dysfunction (e.g., pigment incontinence indicates defective cell to cell transfer of biomolecules). In the pig studies, it has been observed that doses of radiation primarily to the skin have adverse effects that would not be expected in internal organs (e.g., heart, lungs, white blood cells in the circulating blood, spleen, and bone marrow). At doses of 25 cGy to 2 Gy of SPE proton radiation, ferrets had increased bleeding times beginning shortly after irradiation. By 13 days after receiving a dose of 2 Gy SPE proton, the ferrets had severe clotting abnormalities and many of the irradiated ferrets developed symptoms of disseminated intravascular coagulation (DIC). Mice exposed to SPE proton radiation exhibited breaks in the intestinal epithelial barrier, which could allow the entry of bacteria and bacterial products into the circulation and their dissemination in the body. A product of Gram negative bacteria in the gastrointestinal tract, lipopolysaccharide (LPS), was measured in the intervillous space in the terminal ileum in irradiated and hindlimb suspended mice; considerably greater amounts of LPS could be measured in the intervillous space in mice exposed to SPE proton radiation in addition to hindlimb suspension. In a strain of mice considered very sensitive to DNA damaging agents (BALB/c), mice exposed to a dose of 2 Gy SPE proton radiation with or without additional exposure to hindlimb suspension exhibited 100% morbidity (and required euthanasia) when the mice were challenged with a non-toxic bacterial dose (Pseudomonas aeruginosa). Similar trends were observed in outbred mice (ICR mice), but the findings were not as severe. For SPE radiation effects on blood cell counts, the findings in mice, ferrets and pigs were not comparable. The relative biological effectiveness (RBE) values were very different in the three species in which blood cell counts were measured after animal exposure to SPE radiation. The RBE values measured in ferrets and pigs were considerably larger than those calculated for mice. For many endpoints in these studies, the RBE values increased with lower doses of SPE radiation in the dose ranges evaluated.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
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